

Airline Customer Satisfaction & Churn Analysis

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1. Introduction & Business Context

Airline customer satisfaction is no longer a soft metric - it is a leading indicator of churn, repeat purchase behavior, brand equity, and long-term profitability. In an industry characterized by high fixed costs, limited product differentiation, and intense competition, retaining existing customers is often significantly more cost-effective than acquiring new ones.

This report analyzes airline customer satisfaction using survey-based operational and experiential data. The goal extends beyond predicting satisfaction accurately; instead, the emphasis is on understanding why customers become dissatisfied, which service levers matter most, and how airlines should prioritize investments today to protect revenue tomorrow.

Much like advanced player evaluation in professional sports, customer satisfaction must be evaluated holistically. Individual features (such as seat comfort or inflight entertainment) do not operate in isolation. Instead, they interact to form an overall experience that determines whether a customer remains loyal or defects.

2. Dataset Overview & Preparation

The dataset consists of survey responses from nearly 130,000 airline customers, capturing a wide range of attributes including:

- Service quality measures: inflight entertainment, seat comfort, online booking, cleanliness, food and drink, baggage handling
- Operational variables: delays, flight distance, travel class
- Customer characteristics: age, gender, travel type
- Target variable: customer satisfaction (Satisfied vs. Neutral/Dissatisfied)

Data preprocessing included removal of incomplete records, encoding categorical variables, and train/test splitting for out-of-sample evaluation. This ensures the models reflect real performance drivers rather than statistical noise.

3. Modeling Approach & Rationale

Two supervised learning models were selected:

- Decision Tree Classifier: Strength: interpretability; Business value: reveals explicit decision paths that explain satisfaction outcomes

- Random Forest Classifier: Strength: robustness and generalization; Business value: validates findings at scale and reduces sensitivity to noise

Using both models balances explainability with predictive stability. When different approaches converge on the same conclusions, confidence in recommendations increases.

4. Model Performance & Interpretation

Decision Tree: F1-score ~ 94%

Random Forest: F1-score ~ 95%

The small difference indicates that customer satisfaction patterns are clear and well-defined. Key implications:

- Drivers of dissatisfaction are consistent
- Major gains come from service improvements rather than model complexity. Leadership should focus on adopting insights rather than refining models.

5. Decision Tree Insights

The decision tree highlights key features influencing satisfaction. Early splits are strategically important:

- Inflight Entertainment Quality
- Seat Comfort
- Ease of Online Booking

Early splits represent high-leverage interventions. Improving these impacts thousands of customers simultaneously. If these core experiences fail, downstream improvements rarely compensate.

6. Feature Importance

- Inflight entertainment and seat comfort dominate feature importance
- Online booking shapes pre-flight expectations
- Operational variables (flight distance, delays) matter but less than experience quality

Insight: Customers judge airlines more by tolerable experience than operational perfection.

7. Random Forest Validation

Random Forest aggregates multiple decision trees, confirming feature importance rankings of the Decision Tree. This ensures findings are robust, not overfitting artifacts, and reliable across customer segments.

8. Financial & Strategic Impact

Customer satisfaction drives repeat bookings, loyalty program engagement, reduced service recovery costs, and positive word-of-mouth.

Scenario: Improving satisfaction for 2% of dissatisfied customers with \$400 incremental value each yields \$45 million annual upside, excluding secondary benefits.

9. Strategic Recommendations

- Prioritize Experience Over Operations: invest in seat comfort and inflight entertainment for highest return.
- Reduce Digital Friction Early: streamline online booking to improve expectations.
- Embed Predictive Scoring Into Operations: identify at-risk customers post-booking and post-flight for targeted interventions.

10. Forward-Looking Analytics Roadmap

- Integrate time-based and seasonal features
- Incorporate Customer Lifetime Value (CLV) into prioritization
- Test gradient boosting models for incremental insights
- Build executive dashboards linking satisfaction drivers to revenue and churn

11. Conclusion

Airline customer satisfaction is predictable and actionable. Most influential drivers are experiential, not operational or demographic.

Strategy must move beyond punctuality metrics toward holistic experience design. The opportunity is not better models but better decisions powered by data.